

Supplemental Online Content

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eFigure 1. Time-Dependent ROC Curve Analysis of Different Recurrence Prediction Models

eFigure 2. Kaplan-Meier Curves for AF Recurrence in the Ablation Arm

eFigure 3. Time-Dependent ROC Curve Analysis of Different Recurrence Prediction Models in Patients With Echocardiograms

eFigure 4. Propensity Score–Weighted Hazard Ratios of End Points by Intention-to-Treat

eFigure 5. Adjusted Hazard Ratios of End Points by Intention-to-Treat Among Patients With Different Numbers of NMRRFs

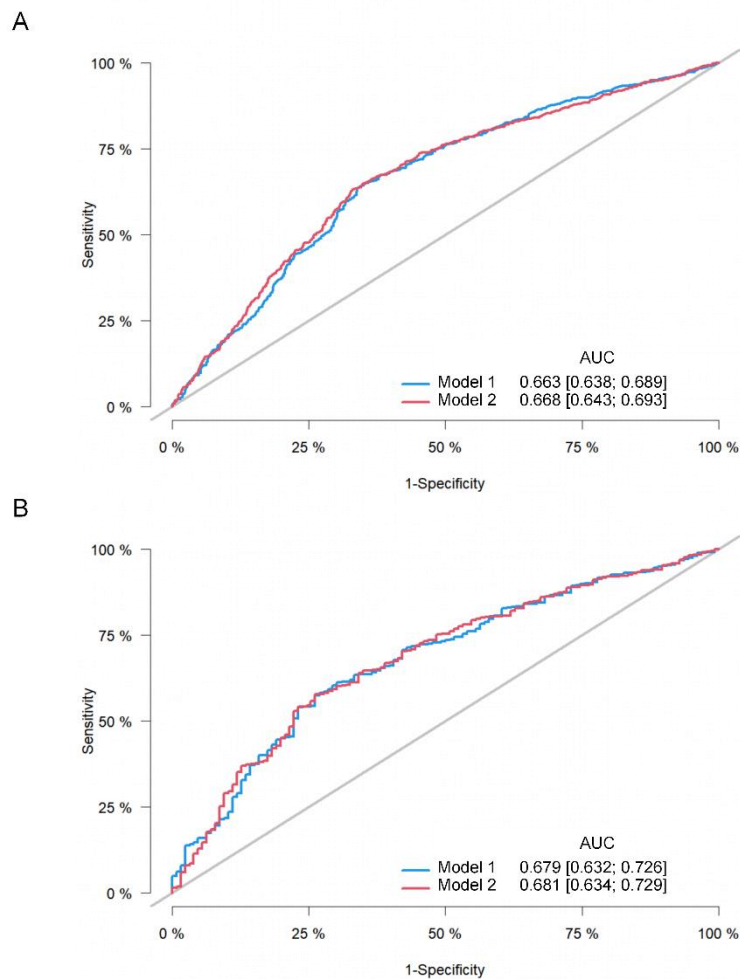
eFigure 6. Propensity Score–Weighted Hazard Ratios of End Points by Intention-to-Treat Among Patients With Different Numbers of NMRRFs

eFigure 7. Quality-of-Life Outcomes by MAFSI Severity Scoring

eTable 1. Emulated Target Trial Framework for Prognosis Among Patients With AF and Fewer Than 3 NMRRFs

eTable 2. Emulated Target Trial Framework for Prognosis Among Patients With AF and 3 or More NMRRFs

This supplemental material has been provided by the authors to give readers additional information about their work.



eFigure 1. Time-Dependent ROC Curve Analysis of Different Recurrence Prediction Models

A: ROC curve of AF recurrence in 12 months. B: ROC curve of AF recurrence in 48 months.

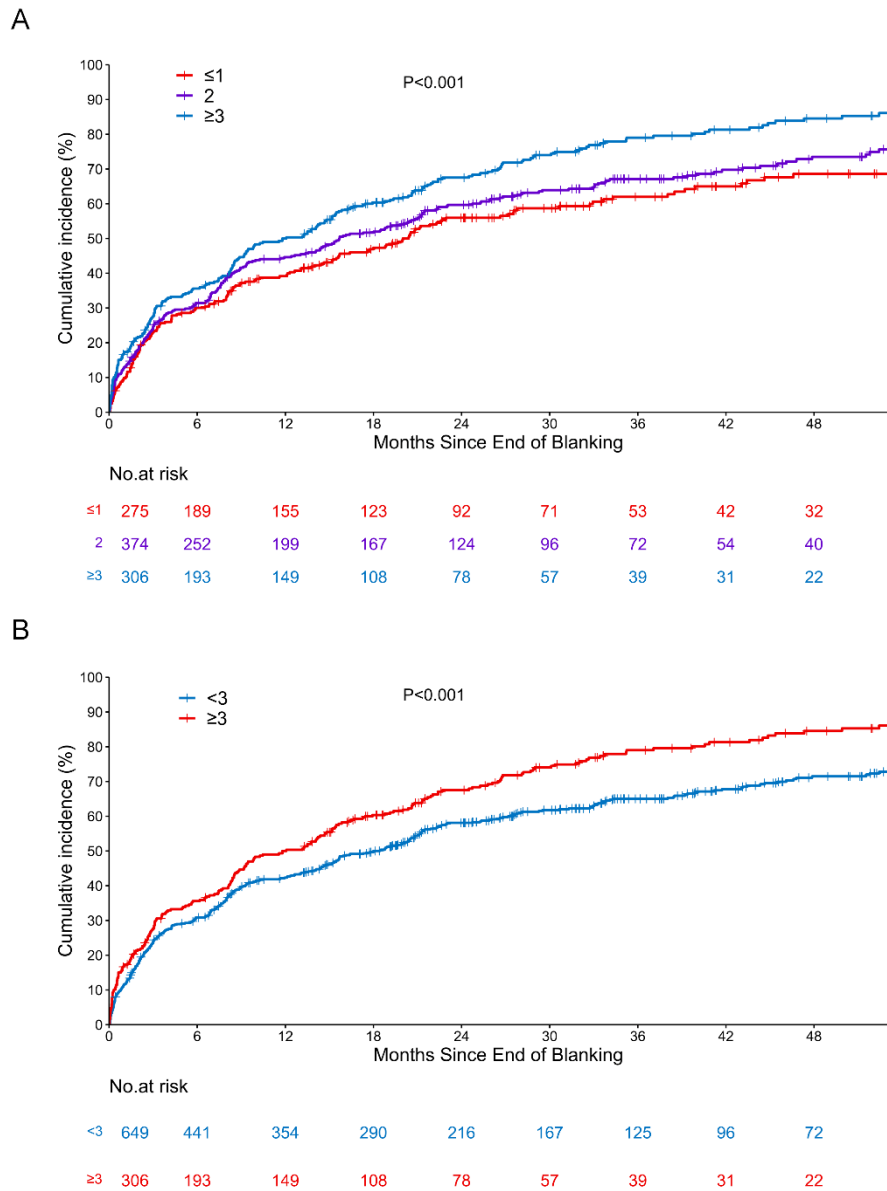
Model 1 includes the treatment group and four NMRRFs (AF duration >1 year, persistent or long-standing persistent atrial fibrillation, age >65 years, and female sex).

Model 2 includes modifiable risk factors (body mass index, alcohol consumption, smoking, diabetes, hypertension, heart failure, and sleep apnea) and all factors in Model 1.

AF=atrial fibrillation, AUC=area under the curve, ROC=receiver operating characteristic,

NMRRFs=non-modifiable recurrence risk factors.

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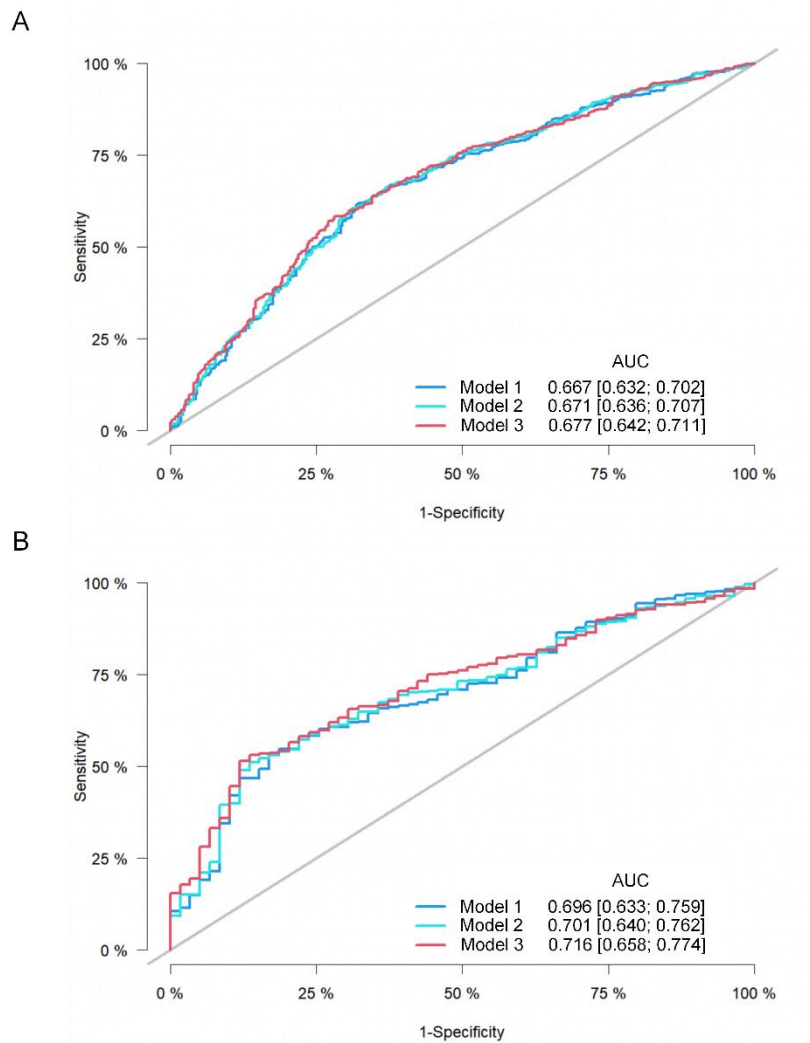
eFigure 2. Kaplan-Meier Curves for AF Recurrence in the Ablation Arm

A: Kaplan–Meier curves for AF recurrence among patients with different numbers of NMRRFs.

B: Kaplan–Meier curves for AF recurrence among patients with fewer than three NMRRFs and patients with three or more NMRRFs.

Non-modifiable recurrence risk factors included AF duration >1 year, persistent or long-standing persistent atrial fibrillation, age >65 years, and female sex.

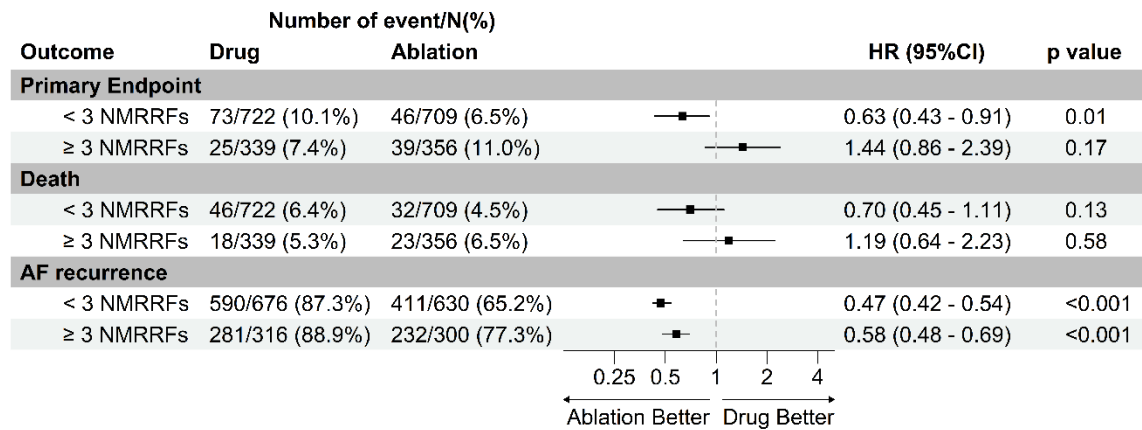
AF=atrial fibrillation, NMRRFs=non-modifiable recurrence risk factors.



eFigure 3. Time-Dependent ROC Curve Analysis of Different Recurrence Prediction Models in Patients With Echocardiograms

A: ROC curve of AF recurrence in 12 months. B: ROC curve of AF recurrence in 48 months.

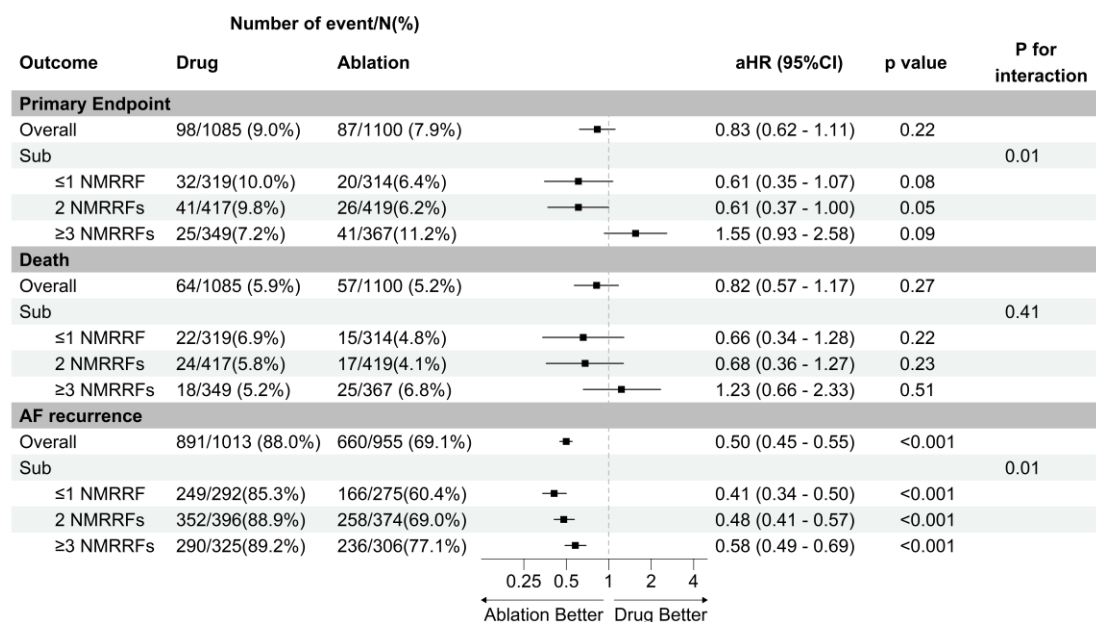
Model 1 includes treatment group, AF duration >1 year, persistent or long-standing persistent atrial fibrillation, age >65 years, and female sex. Model 2 includes left atrial diameter >45 mm and all factors in Model 1. Model 3 includes modifiable risk factors (body mass index, alcohol consumption, smoking, diabetes, hypertension, heart failure, and sleep apnea) and all factors in Model 2.



eFigure 4. Propensity Score–Weighted Hazard Ratios of End Points by Intention-to-Treat

Non-modifiable recurrence risk factors included AF duration >1 year, persistent or long-standing persistent atrial fibrillation, age >65 years, and female sex.

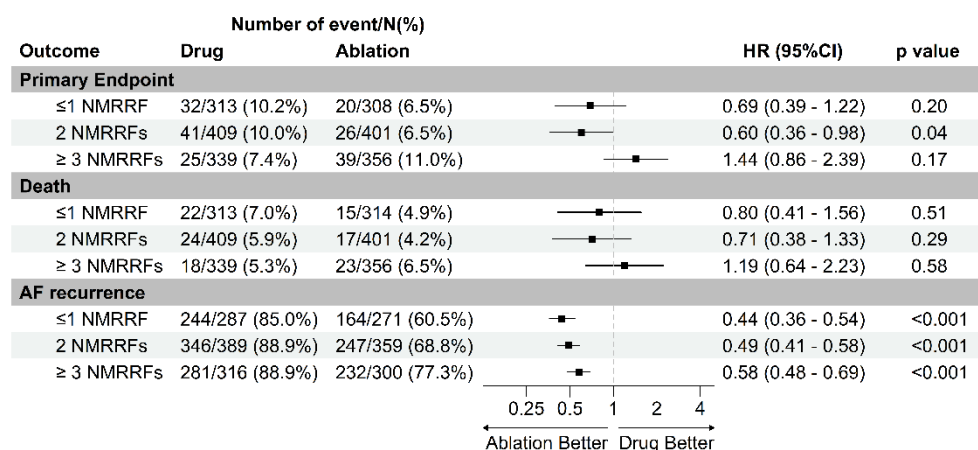
AF=atrial fibrillation, HR= hazard ratio, CI=confidence interval, NMRRFs=non-modifiable recurrence risk factors.



eFigure 5. Adjusted Hazard Ratios of End Points by Intention-to-Treat Among Patients With Different Numbers of NMRRFs

Non-modifiable recurrence risk factors included AF duration >1 year, persistent or long-standing persistent atrial fibrillation, age >65 years, and female sex.

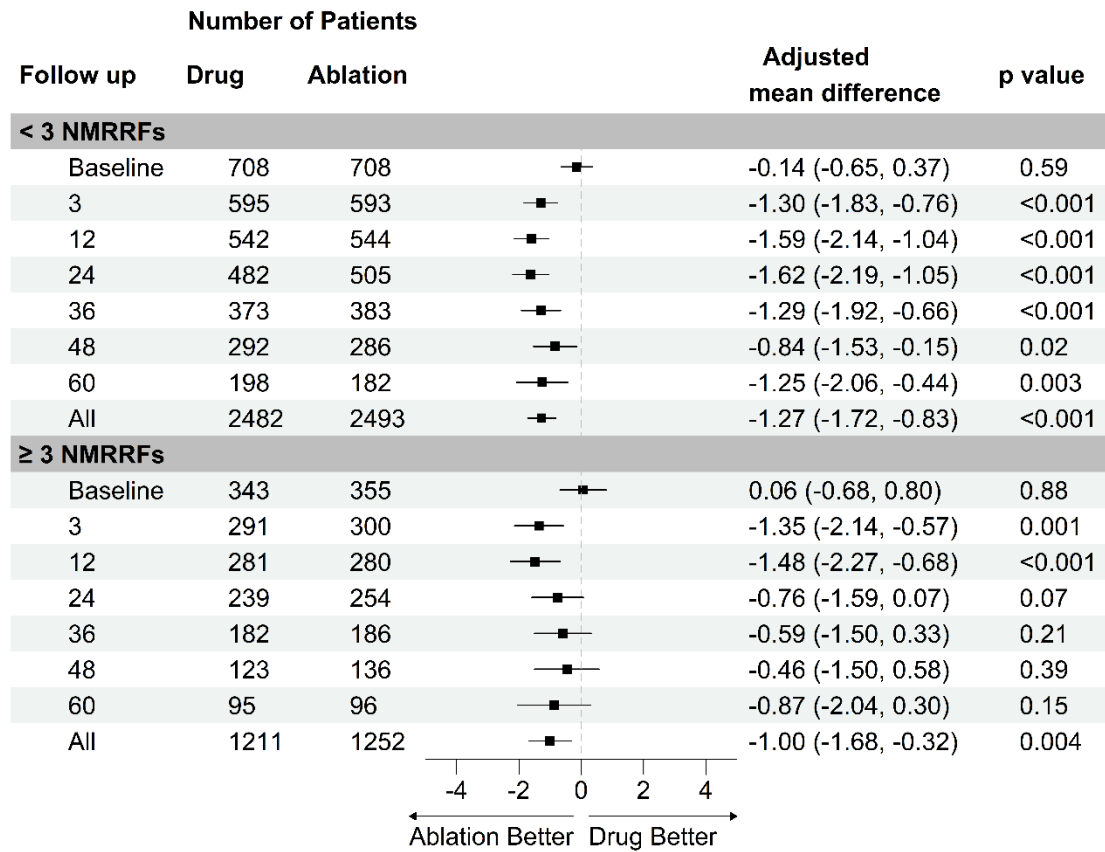
AF=atrial fibrillation, aHR=adjusted hazard ratio, CI=confidence interval, NMRRFs=non-modifiable recurrence risk factors.



eFigure 6. Propensity Score–Weighted Hazard Ratios of End Points by Intention-to-Treat Among Patients With Different Numbers of NMRRFs

Non-modifiable recurrence risk factors included AF duration >1 year, persistent or long-standing persistent atrial fibrillation, age >65 years, and female sex.

AF=atrial fibrillation, HR= hazard ratio, CI=confidence interval, NMRRFs=non-modifiable recurrence risk factors.



eFigure 7. Quality-of-Life Outcomes by MAFSI Severity Scoring

Non-modifiable recurrence risk factors included AF duration >1 year, persistent or long-standing persistent atrial fibrillation, age >65 years, and female sex.

MAFSI=Mayo Atrial Fibrillation-Specific Symptom Inventory, NMRRFs=non-modifiable recurrence risk factors.

eTable 1. Emulated Target Trial Framework for Prognosis Among Patients With AF and Fewer Than 3 NMRRFs

Protocol Component	Target Trial	Emulated Trial
Aim	To evaluate the risk of a composite outcomes (including all-cause mortality, disabling stroke, major bleeding, or cardiac arrest) among patients with AF who have fewer than three NMRRFs.	Same as for the target trial.
Eligibility Criteria	● Aged 65 years and older or younger than 65 years with 1 or more risk factors for stroke (hypertension, heart failure, history of stroke, diabetes, or other heart problems). ● Had 2 or more episodes of paroxysmal AF or 1 episode of persistent AF in the prior 6 months. ● Suitable for catheter-based treatment or rhythm and/or rate control drug therapy. ● Had fewer than 3 NMRRFs (AF duration >1 year, persistent or long-standing persistent AF, age > 65, and female sex). ● Without a prior left atrial catheter ablation for AF. ● Without a history of failure of two or more antiarrhythmic drugs.	Same as for the target trial.
Treatment Strategies	1. Catheter ablation Ablation procedures all included pulmonary vein isolation. The addition of ancillary ablation techniques, were left to the discretion of the investigators. 2. Drug therapy Patients randomized to drug therapy receive rate control medications first. If the patient had previously failed rate control therapy, then rhythm control drug therapy could be initiated in an approach consistent with contemporaneous guidelines.	Same as for the target trial.
Treatment Assignment	Individuals are randomly assigned and are aware of the treatment to which they have been assigned.	1. The CABANA trial did not use stratified randomization based on the number of NMRRFs. But baseline characteristics were similar between the ablation and drug therapy groups. 2. The results were adjusted for a pre-specified set of baseline patient characteristics (age, sex, race, AF

Protocol Component	Target Trial	Emulated Trial
		type, AF duration, history of heart failure, structural heart disease, CHA₂DS₂-VASc score, history of coronary artery disease, and hypertension). 3. We also used propensity score weighting (overlap weighting) to strengthen covariate balance and address potential confounding.
Follow-up	Starts at the time of grouping and ends on the day of death, disabling stroke, serious bleeding, cardiac arrest, loss of follow-up, or the end of the study period (December 31, 2017).	Same as for the target trial.
Outcomes	Primary endpoint: a composite of death, disabling stroke, serious bleeding, or cardiac arrest. Secondary endpoints: all-cause mortality, recurrence of atrial fibrillation.	Same as for the target trial.
Causal Contrasts	Intention-to-treat effect.	Same as for the target trial.
Statistical Analysis	Comparison of the rates and adjusted hazard ratios, among individuals assigned to each treatment strategy.	Same as for the target trial.

AF=atrial fibrillation. NMRRFS=non-modifiable recurrence risk factors.

eTable 2. Emulated Target Trial Framework for Prognosis Among Patients With AF and 3 or More NMRRFs

Protocol Component	Target Trial	Emulated Trial
Aim	To evaluate the risk of a composite outcomes (including all-cause mortality, disabling stroke, major bleeding, or cardiac arrest) among patients with AF who have three or more NMRRFs.	Same as for the target trial.
Eligibility Criteria	● Aged 65 years and older or younger than 65 years with 1 or more risk factors for stroke (hypertension, heart failure, history of stroke, diabetes, or other heart problems). ● Had 2 or more episodes of paroxysmal AF or 1 episode of persistent AF in the prior 6 months. ● Suitable for catheter-based treatment or rhythm and/or rate control drug therapy. ● Had three or more NMRRFs (AF duration >1 year, persistent or long-standing persistent AF, age > 65, and female sex). ● Without a prior left atrial catheter ablation for AF. ● Without a history of failure of two or more antiarrhythmic drugs.	Same as for the target trial.
Treatment Strategies	1. Catheter ablation Ablation procedures all included pulmonary vein isolation. The addition of ancillary ablation techniques were left to the discretion of the investigators. 2. Drug therapy Patients randomized to drug therapy receive rate control medications first. If the patient had previously failed rate control therapy, then rhythm control drug therapy could be initiated in an approach consistent with contemporaneous guidelines.	Same as for the target trial.
Treatment Assignment	Individuals are randomly assigned and are aware of the treatment to which they have been assigned.	1. The CABANA trial did not use stratified randomization based on the number of NMRRFs. But baseline characteristics were similar between the ablation and drug therapy groups. 2. The results were adjusted for a pre-specified set of baseline patient characteristics (age, sex, race, AF type, AF duration, history of heart failure, structural heart disease,

Protocol Component	Target Trial	Emulated Trial
		CHA ₂ DS ₂ -VASc score, history of coronary artery disease, and hypertension). 3. We also used propensity score weighting (overlap weighting) to further strengthen covariate balance and address potential confounding.
Follow-up	Starts at the time of grouping and ends on the day of death, disabling stroke, serious bleeding, cardiac arrest, loss of follow-up, or the end of the study period (December 31, 2017).	Same as for the target trial.
Outcomes	Primary endpoint: a composite of death, disabling stroke, serious bleeding, or cardiac arrest. Secondary endpoints: all-cause mortality, recurrence of atrial fibrillation.	Same as for the target trial.
Causal Contrasts	Intention-to-treat effect.	Same as for the target trial.
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AF=atrial fibrillation. NMRRFS=non-modifiable recurrence risk factors.